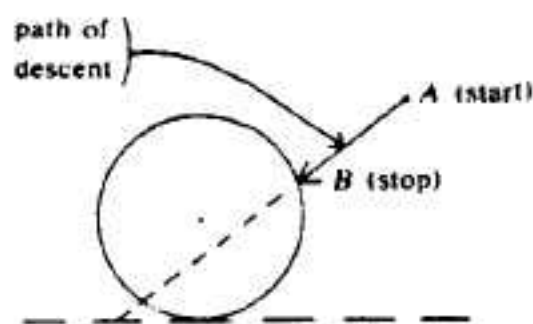


**THE THIRTY-FIFTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

December 7, 1974

- A-1. Call a set of positive integers "conspiratorial" if no three of them are pairwise relatively prime. (A set of integers is "pairwise relatively prime" if no pair of them has a common divisor greater than 1.) What is the largest number of elements in any "conspiratorial" subset of the integers 1 through 16?
- A-2. A circle stands in a plane perpendicular to the ground and a point A lies in this plane exterior to the circle and higher than its bottom. A particle starting from rest at A slides without friction down an inclined straight line until it reaches the circle. Which straight line allows descent in the shortest time? [Assume that the force of gravity is constant over the region involved, there are no relativistic effects, etc.]



The starting point A and the circle are fixed; the stopping point B is allowed to vary over the circle.

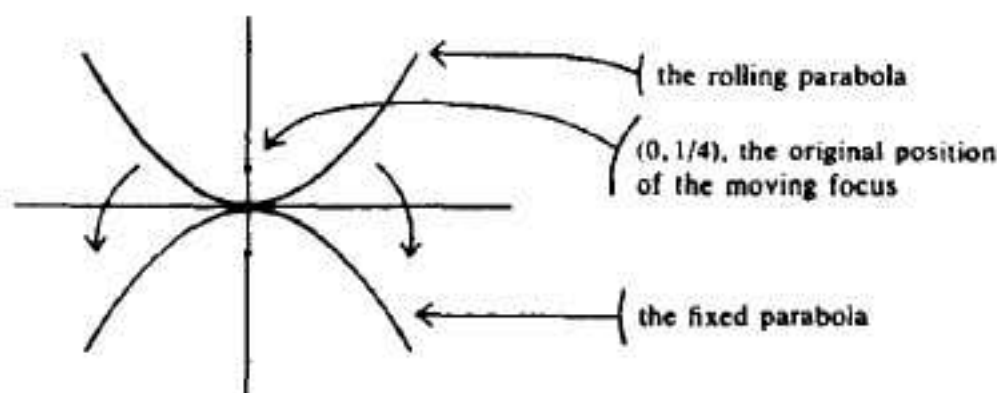
Note. The answer may be given in any form which specifies the line of descent in an unambiguous manner; it is not required to find the coordinates of the point B .

- A-3. A well-known theorem asserts that a prime $p > 2$ can be written as the sum of two perfect squares ($p = m^2 + n^2$, with m and n integers) if and only if $p \equiv 1 \pmod{4}$. Assuming this result, find which primes $p > 2$ can be written in each of the following forms, using (not necessarily positive) integers x and y :
- (a) $x^2 + 16y^2$;
(b) $4x^2 + 4xy + 5y^2$.
- A-4. An unbiased coin is tossed n times. What is the expected value of $|H - T|$, where H is the number of heads and T is the number of tails? In other words, evaluate in *closed form*:

$$\frac{1}{2^{n-1}} \sum_{k=0}^n (n - 2k) \binom{n}{k}.$$

(In this problem, "closed form" means a form not involving a series. The given series can be reduced to a single term involving only binomial coefficients, rational functions of n and 2^n , and the greatest integer function $\{x\}$.)

- A-5. Consider the two mutually tangent parabolas $y = x^2$ and $y = -x^2$. [These have foci at $(0, 1/4)$ and $(0, -1/4)$, and directrices $y = -1/4$ and $y = 1/4$, respectively.] The upper parabola rolls without slipping around the fixed lower parabola. Find the locus of the focus of the moving parabola.



- A-6. It is well known that the value of the polynomial $(x+1)(x+2)\cdots(x+n)$ is exactly divisible by n for every integer x . Given n , let $k = k(n)$ be the *minimal degree* of any monic integral polynomial

$$f(x) = x^k + a_1x^{k-1} + \cdots + a_k$$

(with integer coefficients and leading coefficient 1) such that the value of $f(x)$ is exactly divisible by n for every integer x .

Find the relationship between n and $k = k(n)$. In particular, find the value of k corresponding to $n = 1\,000\,000$.

- B-1. Which configurations of five (not necessarily distinct) points $p_1, \dots, p_5$ on the circle $x^2 + y^2 = 1$ maximize the sum of the ten distances

$$\sum_{i < j} d(p_i, p_j)?$$

[Here $d(p, q)$ denotes the straight line distance between p and q .]

- B-2. Let $y(x)$ be a continuously differentiable real-valued function of a real variable x . Show that if $(y')^2 + y^3 \rightarrow 0$ as $x \rightarrow +\infty$, then $y(x)$ and $y'(x) \rightarrow 0$ as $x \rightarrow +\infty$.

- B-3. Prove that if α is a real number such that

$$\cos \pi\alpha = 1/3,$$

then α is irrational. (The angle $\pi\alpha$ is in radians.)

- B-4. In the standard definition, a real-valued function of two real variables $g: R^2 \rightarrow R^1$ is *continuous* if, for every point $(x_0, y_0) \in R^2$ and every $\varepsilon > 0$, there is a corresponding $\delta > 0$ such that $\{(x - x_0)^2 + (y - y_0)^2\}^{1/2} < \delta$ implies $|g(x, y) - g(x_0, y_0)| < \varepsilon$.

By contrast, $f: R^2 \rightarrow R^1$ is said to be *continuous in each variable separately* if, for each fixed value y_0 of y , the function $f(x, y_0)$ is continuous in the usual sense as a function of x , and similarly $f(x_0, y)$ is continuous as a function of y for each fixed x_0 .

Let $f: R^2 \rightarrow R^1$ be continuous in each variable separately. Show that there exists a sequence of continuous functions $g_n: R^2 \rightarrow R^1$ such that

$$f(x, y) = \lim_{n \rightarrow \infty} g_n(x, y) \text{ for all } (x, y) \in R^2.$$

- B-5. Show that $1 + (n/1!) + (n^2/2!) + \cdots + (n^n/n!) > e^n/2$ for every integer $n \geq 0$.

REMARKS. You may assume as known Taylor's remainder formula:

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} = \frac{1}{n!} \int_0^x (x-t)^n e^t dt,$$

as well as the fact that

$$n! = \int_0^\infty t^n e^{-t} dt.$$

- B-6. For a set with n elements, how many subsets are there whose cardinality (the number of elements in the subset) is respectively $\equiv 0 \pmod{3}$, $\equiv 1 \pmod{3}$, $\equiv 2 \pmod{3}$? In other words, calculate

$$s_{i,n} = \sum_{k \equiv i \pmod{3}} \binom{n}{k} \quad \text{for } i = 0, 1, 2.$$

Your result should be strong enough to permit direct evaluation of the numbers $s_{i,n}$ and to show clearly the relationship of $s_{0,n}$ and $s_{1,n}$ and $s_{2,n}$ to each other for all positive integers n . In particular, show the relationships among these three sums for $n = 1000$. [An illustration of the definition of $s_{i,n}$ is

$$s_{0,6} = \binom{6}{0} + \binom{6}{3} + \binom{6}{6} = 22.]$$